

The Third Dimension

What drives the cogs of 3D? Read on find out the latest and the future in 3D technology



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A standard TV set displays images and video in two dimensions (2D). It lacks the depth that forms the third dimension of viewing.

In real life, when you view an object, either eye sees a slightly different picture from the other due to a minute variation in the angle. The difference between the views of both eyes is greatest for objects closeup and tapers off for those farther away. Using that information your brain calculates the distance between you and the object, helping you perceive depth and see in three dimensions or 3D.

To create a sense of depth on the screen in 3D, videos are shot from two slightly different angles correspond-

ing to the average distance between human eyes. In animated films, these 'shots' come from computer models that generate the two views. In live action, two cameras are used to record stereo video.

At the movie theatre: polarisation

Previously, the 3D effect was created by using colour to separate the image intended for either eye. It utilised the anaglyph method of encoding a 3D image in a single picture by superimposing a pair of pictures.

Current technology utilises a property of light known as 'polarisation.' In a cinema hall, the 3D projector uses polarised filters to project two images—a right-eye perspective displayed with clockwise-polarised

light and a left-eye perspective with counterclockwise light. The audience wears special polarised glasses that allow only the right-polarity light to enter each eye. The brain receives the two different perspectives with different polarisations and assembles these to create an image that has depth.

If polarisation is linear, the audience have to sit with their faces aligned to the screen. But if circular polarisation is used, the audience are free to sit in a more comfortable position.

At home: active and passive 3D

Active 3D. 3D-enabled LCD or plasma TV sets work quite differently from the technology used in cinema halls. TV-based 3D technology rapidly flashes alternating left-eye and right-eye video frames. The glasses worn by the viewer are also not polarised but active shutter glasses.

As a right-eye video frame flashes on the screen, the LCD over the right eye switches from opaque to transparent state. When the left-eye video frame appears, the right-eye LCD turns opaque again and the left-eye LCD becomes transparent. At any moment, you see only one perspective, through one eye. But the left and right video images alternate so quickly—at 120 hertz (times per second)—that you perceive a full 3D view. This illusion is possible due to persistence of vision.

What is not so great about this technology is that you have to wear glasses.

Passive 3D. Here the TV divides left- and right-eye perspectives into alternating vertical columns. Microscopic lenses over the screen 'bend' the light so that slices of the right-side perspective reach the viewer's right